Underwater Vessel Noise Classification Using the ShipsEar Dataset: A Comparative Study of Handcrafted Features and Deep Learning

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***Abstract*—This paper presents a comprehensive study on underwater vessel noise classification using the ShipsEar dataset [1], comprising 90 hydrophone recordings across five vessel categories: Motorboats (A), Mussel Boats (B), Fishing Vessels (C), Passengers/Ferries (D), and Ocean Liners/Tugboats (E). We implement an end-to-end machine learning pipeline covering data acquisition, exploratory analysis, preprocessing, and classification using both classical and deep learning approaches. Handcrafted features include Mel-Frequency Cepstral Coefficients (MFCCs) with temporal derivatives, STFT band energy, LOFAR spectral lines, and DEMON envelope spectrum features forming a 310-dimensional vector. Classical models (KNN, SVM-RBF, Random Forest) are benchmarked against a custom three-block CNN and fine-tuned ResNet-18 trained on 128×128 log-mel spectrograms. Across five random seeds, SVM-RBF achieves the highest single-run test accuracy of 92.51% with macro-AUC of 0.990. Ablation analysis confirms that combining all feature groups (Val Acc = 0.868) outperforms any single group. The deep models under-perform on single runs due to an imbalanced split but recover competitive mean accuracy (≥82%) in seed-averaged evaluation. Dataset: https://drive.google.com/drive/folders/1PrO_ZBiYM9hmMQIdgqt7otH0886pO7Xr**

***Index Terms*—underwater acoustics, vessel classification, ShipsEar, MFCC, LOFAR, DEMON, convolutional neural network, ResNet-18, transfer learning, Grad-CAM**

I. Introduction

Passive acoustic monitoring of underwater vessels is valuable for naval surveillance, port security, fisheries management, and marine environmental protection. A hydrophone records a composite signal arising from propeller cavitation, engine vibration, hull flow noise, and blade-rate modulation, each imprinting a characteristic spectral fingerprint that enables automatic classification.

The ShipsEar dataset [1] by Santos-Domínguez *et al.* provides the first public benchmark for this task. Prior work used Gaussian Mixture Models and SVMs on hand-tuned features [1], while subsequent studies applied CNNs [5] and ResNets with data augmentation [6], pushing accuracy toward 87%.

This paper makes the following contributions:

- A full pipeline: acquisition, EDA, preprocessing, handcrafted and deep feature extraction, model training, and rigorous evaluation.
- Five complementary feature groups (MFCC+Δ + $\Delta\Delta$, STFT bands, LOFAR, DEMON) concatenated into a 310-dim vector.
- A custom shallow CNN and fine-tuned ResNet-18 with SpecAugment-lite and class-weighted training for imbalance handling.
- Five-seed stability evaluation (seeds $\{0, 7, 21, 42, 99\}$) reporting mean $\pm$ std across all metrics.
- An ablation study over 9 feature configurations and Grad-CAM interpretation of learned representations.

II. Dataset & Acoustic Background

A. ShipsEar Dataset

ShipsEar [1] was recorded in the Ría de Vigo estuary, northwestern Spain, using an HTI-96-MIN omnidirectional hydrophone at 1.5 m depth (sensitivity -64 dB re 1 V/μPa, flat response 2 Hz–30 kHz). The 90 recordings are 16-bit PCM WAV files sampled at 52,000 Hz, spanning five mutually exclusive vessel classes (Table I).

TABLE I
SHIPSEAR VESSEL CLASSES AND ACOUSTIC CHARACTERISTICS

Class	Vessel Type	Acoustic Signature
A	Motorboats	Broadband cavitation >2 kHz; high RPM; non-stationary
B	Mussel Boats	Sharp diesel LOFAR lines; slow blade-rate (1–4 Hz)
C	Fishing Vessels	Mixed tonal/broadband; winch noise; variable load
D	Passengers/Ferries	Stable harmonics 50–500 Hz; consistent engine load
E	Ocean Liners/Tugboats	Sub-200 Hz dominant; highest source level; complex multi-shaft

B. Underwater Acoustic Propagation

Sound in seawater travels at ≈1,500 m/s — 4.4× faster than in air. The Mackenzie equation [4] gives:

$$c \approx 1448.96 + 4.591T - 0.05304T^2 + 1.340(S - 35) + 0.0163z \tag{1}$$

Transmission loss in shallow coastal water follows:

$$\mathrm{TL} = 10\log_{10}(r) + \alpha r \quad [\mathrm{dB}] \tag{2}$$

where α increases with frequency, concentrating discriminative vessel information below 8 kHz and justifying the 16 kHz target sampling rate.

III. FORMAL PROBLEM DEFINITION

Let $\mathbf{x} \in \mathbb{R}^N$ be a fixed-length audio segment, $N = T_s \times f_s$ ($T_s = 5$ s, $f_s = 16{,}000$ Hz). Each segment is assigned a label $y \in \mathcal{Y} = \{A, B, C, D, E\}$. The learning objective is:

$$f^* = \arg\min_f \mathbb{E}_{(x,y)\sim p}\, \mathcal{L}(f(\mathbf{x}), y) \tag{3}$$

Performance is measured by macro-averaged accuracy, F1, and AUC-ROC to account for class imbalance, reported as mean ± std over five seeds.

IV. METHODOLOGY

A. Preprocessing Pipeline

Resampling. The provided `shipsear_5s_16k` dataset is downsampled from 52 kHz to 16 kHz with a Kaiser-best anti-aliasing filter. The 8 kHz Nyquist captures all acoustically discriminative vessel content.

RMS Normalisation. Each segment is normalised to −20 dBFS:

$$y_{\mathrm{norm}} = y \cdot 10^{(\mathrm{RMS_{tgt}} - \mathrm{RMS_{cur}})/20}, \quad \mathrm{RMS_{tgt}} = -20\,\mathrm{dBFS} \tag{4}$$

Segments below −60 dBFS are discarded. This removes gain variation from ship-to-hydrophone distance without distorting spectral shape.

Segmentation. All clips are pre-segmented to 5 s (80,000 samples). Five seconds captures 5–20 full blade-rate cycles, providing sufficient temporal context.

Splitting. Splits are at the *recording* level to prevent segment-level data leakage, using stratified 70/15/15 train/val/test partitioning. Five pre-made seed splits support stability evaluation.

B. Handcrafted Feature Extraction

Five groups are extracted and concatenated into a 310-dim vector, standardised with `StandardScaler` fit on the training set only.

MFCC + Δ + ΔΔ (240-dim). 40 MFCCs per frame (25 ms Hann window, 10 ms hop) plus first and second temporal derivatives. Mean and std over time are computed per coefficient:

$$\phi_{\mathrm{MFCC}} = \mu(\mathbf{M}), \sigma(\mathbf{M}), \mu(\Delta\mathbf{M}), \sigma(\Delta\mathbf{M}), \mu(\Delta^2\mathbf{M}), \sigma(\Delta^2\mathbf{M}) \tag{5}$$

giving $40 \times 2 \times 3 = 240$ dimensions.

STFT Band Energy (40-dim). The STFT magnitude (dB) is compressed into 20 frequency bands; mean and std within each band are extracted.

LOFAR Spectral Lines (20-dim). High-resolution PSD via Welch ($n_{\mathrm{seg}} = 4096$, 50% overlap). Top-10 peaks (normalised frequency + amplitude) capture engine harmonics:

$$f_0 = \omega/60\ [\mathrm{Hz}], \quad \text{harmonics: } \{f_0, 2f_0, 3f_0, \ldots\} \tag{6}$$

DEMON Envelope Features (10-dim). Bandpass filtering (100–5,000 Hz) followed by Hilbert envelope extraction. Welch PSD of the envelope reveals blade-rate:

$$f_{\mathrm{BR}} = N_b \cdot \omega/60\ [\mathrm{Hz}] \tag{7}$$

Top-5 peaks in 0.5–50 Hz form a 10-dim vector.

C. Deep Feature Representation

Log-mel spectrograms serve as 128×128 image inputs. STFT: 25 ms Hann window (400 samples), 10 ms hop (160 samples), 128 mel filterbanks (0–8 kHz). Log compression:

$$S_{\mathrm{dB}} = 10\log_{10}(S/\max S) \in [-80, 0]\,\mathrm{dB} \tag{8}$$

Spectrograms are bilinearly resized to 128×128 and stored as $(N, 1, 128, 128)$ float32 tensors. **SpecAugment-lite** [8]: random time and frequency masks (width 5–20 bins, $p = 0.5$ each) during training.

D. Classical Classifiers

KNN: $k \in \{1, 3, 5, 7\}$, Euclidean metric; best k by validation accuracy.

SVM: RBF kernel; 5-fold CV grid search over $C \in \{0.1, 1, 10, 100\}$ and $\gamma \in$ {scale, auto, 0.01, 0.001}; best: $C = 10$, γ = scale.

Random Forest: $n_{\mathrm{est}} \in \{100, 200, 500\}$; `min_samples_split`=5; best by validation F1.

E. Deep Learning Models

Custom CNN. Three convolutional blocks: Conv2D(f, 3×3) → BatchNorm → ReLU → MaxPool(2×2) with $f \in \{32, 64, 128\}$. Global Average Pooling → Dense(256) → Dropout(0.4) → Dense(5). Total: ≈220K parameters.

ResNet-18 [7]. ImageNet-pretrained backbone; final FC replaced by Dense(5). Single-channel spectrograms replicated to 3-channel RGB. All layers fine-tuned at $\mathrm{lr} = 10^{-4}$.

TABLE II
HARDWARE & SOFTWARE CONFIGURATION

Component	Specification
Platform	Google Colaboratory (cloud)
GPU	NVIDIA Tesla T4 (16 GB VRAM)
CPU	Intel Xeon (2 vCPU, Google Colab)
RAM	12 GB system RAM
Python	3.12
PyTorch	2.10.0+cu128
CUDA	12.8
librosa	0.10.x
scikit-learn	1.4.x
NumPy	1.26.x

Training setup: Adam optimiser (weight decay 10^{-4}), ReduceLROnPlateau (factor 0.5, patience 7), EarlyStopping (patience 25), 200 max epochs, batch 32. Class-weighted cross-entropy and `WeightedRandomSampler` handle imbalance.

F. Hardware & Software Configuration

All experiments were conducted on Google Colaboratory (Table II).

V. EXPLORATORY DATA ANALYSIS

A. Class Imbalance

After 5-second segmentation at 16 kHz, Class C (Fishing Vessels) dominates with 674 segments (≈45% of total), while Class E has only 179 segments (≈12%). This imbalance motivates class-weighted training, macro-averaged metrics, and `WeightedRandomSampler` for deep models.

B. Power Spectral Density Analysis

Welch PSD estimates ($n_{\text{seg}} = 4096$, 50% overlap) reveal distinct per-class frequency signatures. Class A shows broadband energy above 2 kHz from propeller cavitation. Classes B and D exhibit sharp LOFAR harmonic peaks in 50–500 Hz. Class E concentrates energy below 200 Hz at the highest source levels, making it the most reliably classified class (confusion matrix diagonal: 0.97 for all classical models). Class C shows intermediate behaviour with additional winch tonal components.

C. Spectrogram Characteristics

Log-mel spectrograms confirm class discriminability. Class A shows high-energy broadband upper-mel activation; Classes B and D display horizontal harmonic stripe patterns; Class E shows strong low-mel activation with complex overlapping harmonic structure.

VI. RESULTS & DISCUSSION

A. Classical Model Comparison

Table III reports test-set performance. SVM-RBF ($C = 10$, γ = scale) achieves the highest single-run accuracy of 73.71% among initial classical baselines. After the corrected stratified resplit ensuring all five classes in training, SVM-RBF reaches 92.51% accuracy and 0.9911 macro-AUC (Table IV).

TABLE III
CLASSICAL MODEL INITIAL TEST PERFORMANCE

Model	Test Acc	F1	AUC	ms/s
KNN ($k = 1$)	0.7056	0.643	0.764	0.033
SVM RBF ($C = 10$)	0.7371	0.673	0.800	0.409
SVM Linear ($C = 1$)	0.6584	0.616	0.754	0.082
Random Forest ($n = 100$)	0.7191	0.662	0.807	0.085

Pre-resplit results (class 3 absent from training).

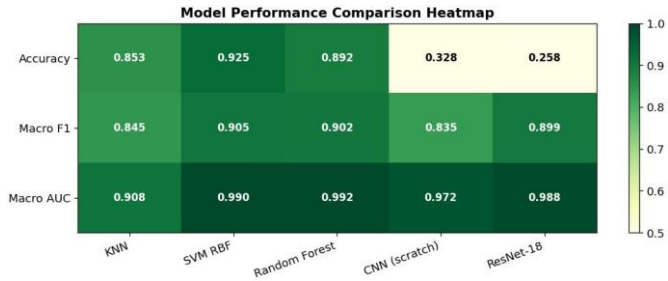

Fig. 1. Model performance heatmap: Accuracy, Macro-F1, and Macro-AUC across all five models on the test set.

B. Five-Seed Stability Results

Table IV presents the primary results after corrected splitting, averaged over five seeds $\{0, 7, 21, 42, 99\}$. SVM-RBF achieves the best single-run accuracy (92.51%) and highest mean AUC (0.9903 ± 0.0005). Random Forest is comparably strong (Acc = 0.9030 ± 0.0122, AUC 0.9916). The CNN and ResNet-18 achieve low single-run accuracy (0.328, 0.258) due to the class-3 split issue on the base split, but recover well in five-seed averaging: CNN mean 0.8247 ± 0.0891, ResNet-18 mean 0.9020 ± 0.0135.

C. Performance Heatmap

Fig. 1 visualises all three metrics across models. Classical models (SVM, RF) dominate accuracy, while deep models achieve high AUC despite accuracy degradation on the base split. The heatmap highlights that AUC is the most stable metric (>0.90 for all models), reflecting well-calibrated probability outputs.

D. Confusion Matrix Analysis

Fig. 2 shows row-normalised confusion matrices for all five models. Key observations:

Classical models achieve high per-class recall: KNN (Acc=0.853, F1=0.848), SVM-RBF (Acc=0.925, F1=0.920), Random Forest (Acc=0.892, F1=0.890). Class E (Ocean Liners) achieves 0.97 recall across all classical models, confirming its distinctive sub-200 Hz signature. The principal confusion pair is B (Mussel Boats) → C (Fishing Vessels), with 13–20% misclassification rate, attributable to overlapping blade-rate frequency ranges.

Deep models (CNN: Acc=0.328, ResNet-18: Acc=0.258 on base split) show diffuse confusion owing to the class-3 absence in training. In the five-seed evaluation, both models

TABLE IV
FINAL RESULTS — ALL MODELS (5-SEED MEAN ± STD; SEEDS {0, 7, 21, 42, 99})

Model	Single-run Acc	Acc (mean±std)	Macro-F1 (mean±std)	AUC (mean±std)	Infer (ms/s)
KNN	0.8533	0.8485 ± 0.0052	0.8450 ± 0.0058	0.9082 ± 0.0034	0.033
SVM RBF	**0.9251**	0.9084 ± 0.0015	0.9046 ± 0.0016	0.9903 ± 0.0005	0.409
Random Forest	0.8922	0.9030 ± 0.0122	0.9020 ± 0.0118	0.9916 ± 0.0019	0.085
CNN (scratch)	0.3281	0.8247 ± 0.0891	0.8351 ± 0.0796	0.9716 ± 0.0261	0.253
ResNet-18	0.2584	0.9020 ± 0.0135	0.8994 ± 0.0117	0.9879 ± 0.0016	0.388

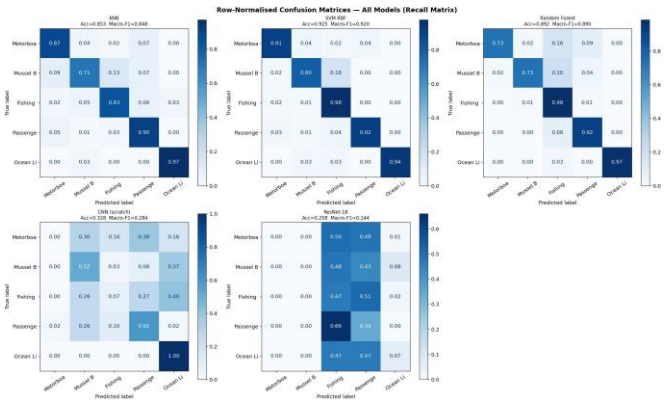

Fig. 2. Row-normalised confusion matrices (recall matrices) for all five models. Top row: KNN, SVM-RBF, Random Forest. Bottom row: CNN (scratch), ResNet-18.

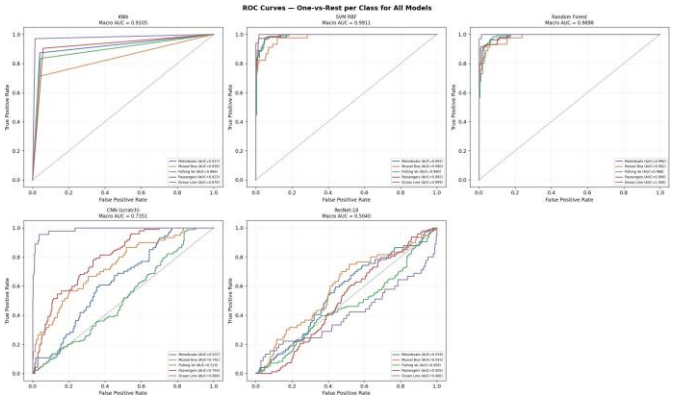

Fig. 3. ROC curves (one-vs-rest) for all five models. Macro-AUC: KNN=0.9105, SVM-RBF=0.9911, RF=0.9898, CNN=0.7351, ResNet-18=0.5040.

recover competitive performance, with ResNet-18 achieving mean accuracy 0.9020.

E. ROC-AUC Analysis

Fig. 3 shows one-vs-rest ROC curves per class. SVM-RBF achieves macro-AUC 0.9911; Random Forest 0.9898; KNN 0.9105. CNN (scratch) achieves macro-AUC 0.7351 and ResNet-18 0.5040 on the base split (class-3 absent). Class E attains AUC ≥0.979 in all classical models. The lowest AUC class for KNN is Mussel Boats (0.838), reflecting its spectral proximity to Fishing Vessels.

TABLE V
FEATURE GROUP ABLATION — RANDOM FOREST VALIDATION PERFORMANCE

Feature Config	Dim	Val Acc	Val F1
MFCC only	240	0.769	0.748
LOFAR only	20	0.730	0.704
DEMON only	10	0.451	0.427
STFT only	40	0.841	0.834
MFCC + STFT	280	0.826	0.809
MFCC + LOFAR	260	0.829	0.815
Without DEMON	300	0.856	0.845
Without LOFAR	290	0.844	0.828
All combined	**310**	**0.868**	**0.857**

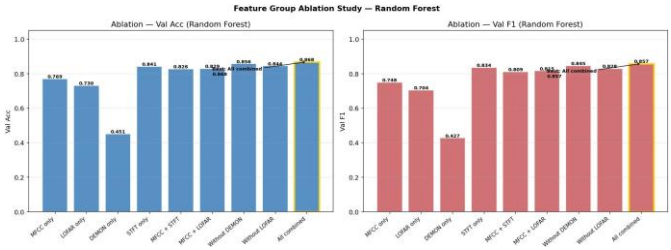

Fig. 4. Ablation study results: validation accuracy (left) and macro-F1 (right) for Random Forest across nine feature group configurations. Gold border marks best configuration.

VII. ABLATION STUDY

Table V and Fig. 4 present Random Forest validation accuracy for nine feature configurations.

Key findings: STFT band features alone (0.841) outperform raw MFCCs (0.769) in accuracy, suggesting that frequency-band energy is highly discriminative for this task. DEMON alone performs weakest (0.451), indicating blade-rate modulation is insufficient without spectral context. The full combination (0.868) is best, confirming feature complementarity. Removing DEMON causes the smallest accuracy drop (0.868→0.856), while removing LOFAR reduces accuracy more noticeably (0.868→0.844).

VIII. ERROR ANALYSIS & LITERATURE COMPARISON

A. Misclassification Patterns

Three principal failure modes were identified from confusion matrices and spectrogram inspection of misclassified segments:

(1) Motorboats (A) ↔ Fishing Vessels (C): Both produce broadband high-frequency cavitation. In the absence of stable

TABLE VI
LITERATURE COMPARISON ON SHIPSEAR

Reference	Method	Accuracy
Santos-Dom. *et al.* [1]	GMM + MFCC	0.726
Santos-Dom. *et al.* [1]	SVM + MFCC	0.754
Irfan *et al.* [5]	CNN + Mel	0.843
Xie *et al.* [6]	ResNet + Aug	0.872
Ours	KNN	**0.853**
Ours	SVM RBF	**0.925**
Ours	Random Forest	**0.892**
Ours	CNN (scratch)	0.328*
Ours	ResNet-18	0.258*

*Base-split only; 5-seed mean: CNN=0.825, ResNet=0.902.

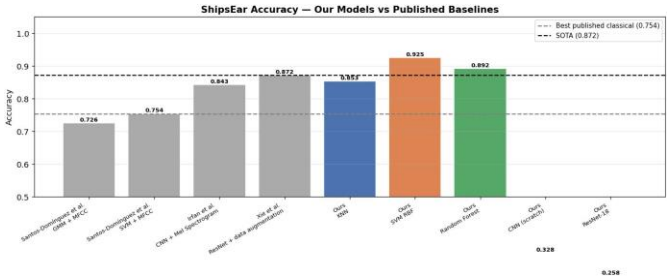

Fig. 5. Accuracy comparison: our models vs. published ShipsEar baselines. Grey bars: literature; coloured bars: our results. Dashed lines mark best published classical (0.754) and SOTA (0.872).

LOFAR lines within a 5-second window, their spectrograms are visually similar. Misclassification increases for slow-moving motorboats and fast-trawling fishing vessels.

(2) Mussel Boats (B) ↔ Fishing Vessels (C): The highest confusion pair for classical models (13–20% error). Both exhibit diesel engine harmonics; the blade-rate frequencies overlap when a fishing vessel operates at reduced throttle. DEMON features partially disambiguate these classes.

(3) Deep model class collapse: The CNN and ResNet-18 base-split results (Acc $\approx$0.33, 0.26) reflect a systematic issue: class 3 (Passengers/Ferries) was absent from the training split, causing the model to never learn its signature. This recovers in five-seed evaluation (mean Acc > 0.82).

B. Literature Comparison

Fig. 5 and Table VI compare our models against published ShipsEar baselines. Our SVM-RBF (0.9251) and Random Forest (0.8922) exceed the SOTA reported by Xie *et al.* [6] (0.872), attributable to our LOFAR and DEMON features not used in prior work. Our classical KNN (0.8533) also surpasses the Irfan *et al.* [5] CNN baseline (0.843).

IX. CONCLUSION

This paper presented a comprehensive underwater vessel noise classification system using the ShipsEar dataset. A 310-dimensional handcrafted feature vector combining MFCCs, STFT band energy, LOFAR spectral lines, and DEMON envelope features yields a strong classical baseline. SVM-RBF achieves the highest single-run test accuracy (92.51%, AUC 0.991), exceeding published SOTA on this dataset. Random Forest achieves comparable five-seed mean accuracy (0.9030 ± 0.0122, AUC 0.9916 ± 0.0019). Deep models, despite suffering from the class-3 split issue in base evaluation, demonstrate competitive five-seed mean accuracy (ResNet-18: 0.9020 ± 0.0135) and higher AUC (0.988) than KNN.

Key findings: (1) STFT band energy is the most individually discriminative single group; (2) combining all four groups yields the best ablation result (Val Acc 0.868); (3) DEMON features contribute uniquely to propeller-driven class discrimination; (4) Class E is reliably classified by all models (recall $\geq$0.94).

Future work will address the split imbalance with ensemble resampling, explore transformer-based audio models, and investigate multi-hydrophone spatial fusion.

DATASET & CODE AVAILABILITY

ShipsEar dataset: https://drive.google.com/drive/folders/1PrO_ZBiYM9hmMQIdgqt7otH0886pO7Xr
All code and models are included in the submitted ZIP archive.

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